Matrix-Free Second-Order Orbital Optimization for Valence Bond Self-Consistent Field Theory

Tao Xia, Suxin Yu, Wei Wu, Chen Zhou*

State Key Laboratory of Physical Chemistry of Solid Surfaces, Fujian Provincial Key Laboratory of Theoretical and Computational Chemistry, and College of Chemistry and Chemical Engineering, Xiamen University, Xiamen, Fujian 361005

*Email: czhou1989@xmu.edu.cn

JCTC Article — Theory draft for review, 7 October 2026

ABSTRACT

Orbital optimization in valence bond self-consistent field (VBSCF) theory must accommodate nonorthogonal orbitals, prescribed atomic orbital (AO) supports, and relaxation of the structure coefficients. We formulate a matrix-free norm-extended optimization (NEO) treatment in support-preserving nonredundant coordinates, retaining orbital–structure coupling through independently expanded orbital and structure response spaces. The numerical study will compare NEO with conventional and block-preconditioned limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) optimization, with truncated Newton as an additional second-order comparison, reporting convergence, operator work, wall time, and peak memory under matched conditions. [Replace this sentence with the principal quantitative findings and the supported conclusion after the benchmark campaign.]

1 INTRODUCTION

1.1 The optimization problem

Introduce valence bond (VB) theory and valence bond self-consistent field (VBSCF) orbital optimization. Explain the use of nonorthogonal localized orbitals and fixed atomic orbital (AO) supports for hybrid atomic orbitals (HAOs). Separate orbital optimization from the structure-coefficient problem. Introduce overlap-enhanced orbitals (OEOs) as full-AO orbitals, not as an orthogonal-orbital model.

1.2 Second-Order Optimization

Review second-order multiconfigurational self-consistent field (MCSCF) methods: restricted-step Newton, augmented-Hessian methods, and norm-extended optimization (NEO), including matrix-free complete active space self-consistent field (CASSCF) implementations [1–3]. Identify what carries over to VBSCF—the local coupled quadratic model and iterative operator actions—and what does not carry over without derivation: orthogonal orbital rotations and their redundancy rules. Introduce limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) optimization as the quasi-Newton comparison.

1.3 Scope of this work

State the methodological contribution as a second-order optimization framework for the specified nonorthogonal VB variational space. Treat integral representation and string contraction as supporting implementations, not independent novelty claims in this paper. The principal numerical question is when additional curvature information improves convergence enough to repay its cost relative to block-preconditioned L-BFGS (block-LBFGS).

2 THEORY

Throughout this section, I, J label VB structures, μ, ν label AO basis functions, i, j label orbitals, n, m label electronic states, and k labels accepted optimization iterations. Boldface denotes matrices and vectors. Calligraphic $\mathcal{H}$ denotes a matrix-free curvature or coupling operator, and square brackets denote its action on a direction. The superscripts o and s identify orbital and structure-coefficient coordinates: the first labels the output space and the second the input space.

2.1 Wave Function and Energy

The valence bond (VB) wave function is written as

$$|\Psi(\mathbf{C}, \mathbf{x})\rangle = \sum_I C_I |\Phi_I(\mathbf{x})\rangle, \tag{1}$$

where $|\Phi_I\rangle$ is the Ith spin-adapted VB structure, $\mathbf{C}$ is the vector of structure coefficients C_I, and $\mathbf{x}$ collects the allowed atomic orbital (AO) coefficients. The sum runs over the chosen structures.

For fixed orbitals, the structure-coefficient vector $\mathbf{C}_n$ of electronic state n is obtained from the generalized eigenvalue problem

$$\mathbf{H}\mathbf{C}_n = E_n \mathbf{M}\mathbf{C}_n, \tag{2}$$

where $\mathbf{H}$ and $\mathbf{M}$ are the electronic Hamiltonian and overlap matrices in the VB structure basis, respectively, and E_n is the electronic energy of state n. The Ith component of $\mathbf{C}_n$, C_{In}, is the coefficient of the Ith VB structure in state n.

In equal-weight state-averaged valence bond self-consistent field (VBSCF) calculations, the orbital objective is

$$\mathcal{E}(\mathbf{x}) = \frac{1}{N_{\text{state}}} \sum_{n=1}^{N_{\text{state}}} E_n(\mathbf{x}), \tag{3}$$

where N_{state} is the number of lowest-energy states included in the average. Orbital optimization minimizes $\mathcal{E}(\mathbf{x})$ with the structure coefficients optimized at each orbital point. The choice $N_{\text{state}} = 1$ recovers ground-state VBSCF.

2.2 Nonredundant Orbital Coordinates

A normalized VB wave function does not have a unique orbital-coefficient representation. Rescaling an orbital leaves its normalized form unchanged, and certain orbital mixings can be compensated by changes of the structure coefficients. These redundant degrees of freedom can make the second-order equations singular or poorly conditioned. We construct nonredundant coordinates by identifying the redundant variations compatible with the orbital localization constraints and retaining a complement to them.

The ith orbital is expressed as a linear combination of atomic orbitals,

$$\phi_i = \sum_\mu b_{\mu i} \chi_\mu, \tag{4}$$

where χ_μ is an AO basis function and $b_{\mu i}$ is its coefficient in orbital i. Hybrid atomic orbitals (HAOs) are strictly localized on a prescribed atom or group of atoms; coefficients outside that region remain zero. Overlap-enhanced orbitals (OEOs) can expand over the entire AO basis without this localization constraint. Let $\mathcal{A}_i$ denote the set of AO indices allowed for orbital i, and let $\mathbf{x}_i = (b_{\mu i})_{\mu \in \mathcal{A}_i}$ collect its variable coefficients. The full parameter vector $\mathbf{x}$ is formed by concatenating these vectors.

Orbital scaling, $\delta\phi_i = \epsilon\phi_i$, is the simplest redundancy, where ϵ is an infinitesimal scalar. Further redundancies arise from changes of basis within the doubly occupied inactive space and from adding inactive components to active orbitals. For a complete active-space expansion with OEOs, changes of basis within the active space can also be compensated by the structure coefficients. This active-space redundancy is not assumed for a selected VB expansion, because mixing active orbitals generally changes the chosen structure space.

More generally, an allowed orbital variation $\delta\mathbf{x}$ is redundant if a structure-coefficient variation $\delta\mathbf{C}$ exists such that

$$D_{\mathbf{x}}|\Psi\rangle[\delta\mathbf{x}] + \sum_I \delta C_I |\Phi_I\rangle = \gamma|\Psi\rangle, \tag{5}$$

where $D_{\mathbf{x}}$ denotes a directional derivative at fixed structure coefficients, δC_I is the Ith component of $\delta\mathbf{C}$, and the scalar γ allows an overall normalization change. The normalized physical state is unchanged to first order. The construction below removes the orbital-representation redundancies described above; it does not classify directions by a small energy curvature.

To construct these directions, let $\mathbf{D}$ contain the AO coefficients of the inactive orbitals. An AO-metric-orthonormal basis of their span and the associated projector are

$$\widetilde{\mathbf{D}} = \mathbf{D}(\mathbf{D}^T\mathbf{S}\mathbf{D})^{-1/2}, \qquad \mathbf{P}_d = \widetilde{\mathbf{D}}\widetilde{\mathbf{D}}^T\mathbf{S}, \tag{6}$$

where $\mathbf{S}$ is the AO overlap matrix, with elements $S_{\mu\nu} = \langle\chi_\mu|\chi_\nu\rangle$, and $\mathbf{P}_d$ projects onto the inactive span in this metric. If there are no inactive orbitals, $\mathbf{P}_d = \mathbf{0}$ and $\widetilde{\mathbf{D}}$ has no columns. Let $\mathbf{A}$ contain the AO coefficients of the current active orbitals and define

$$\mathbf{A}_\perp = (\mathbf{I} - \mathbf{P}_d)\mathbf{A}, \qquad \widetilde{\mathbf{a}}_i = \frac{(\mathbf{A}_\perp)_i}{\sqrt{(\mathbf{A}_\perp)_i^T\mathbf{S}(\mathbf{A}_\perp)_i}}, \qquad \widetilde{\mathbf{A}} = \mathbf{A}_\perp(\mathbf{A}_\perp^T\mathbf{S}\mathbf{A}_\perp)^{-1/2}, \tag{7}$$

where $(\mathbf{A}_\perp)_i$ is the column corresponding to active orbital i, $\widetilde{\mathbf{a}}_i$ is its normalized component outside the inactive span, and $\mathbf{I}$ is the identity matrix. The full basis $\widetilde{\mathbf{A}}$ is needed only when the complete OEO active-space redundancy is used. These orthonormal bases condition the construction of redundant directions; they do not impose mutual orthogonality on the optimized VB orbitals.

The candidate redundant directions for orbital i are the columns of

$$\mathbf{G}_i = \begin{cases} \widetilde{\mathbf{D}}, & i \text{ inactive}, \\ (\widetilde{\mathbf{D}}, \widetilde{\mathbf{a}}_i), & i \text{ active, without the complete OEO gauge}, \\ (\widetilde{\mathbf{D}}, \widetilde{\mathbf{A}}), & i \text{ active, with the complete OEO gauge}. \end{cases} \tag{8}$$

For an inactive orbital, its scaling direction lies in the inactive span. For an active orbital, the inactive span together with its own projected direction contains its scaling direction. The complete OEO case additionally includes mixing with the other active orbitals.

Not every direction in range($\mathbf{G}_i$) respects strict localization. Let $\mathbf{F}_i$ extract the AO components outside $\mathcal{A}_i$. A redundant variation $\mathbf{G}_i\mathbf{t}$ is allowed only if

$$\mathbf{F}_i\mathbf{G}_i\mathbf{t} = \mathbf{0}. \tag{9}$$

Geometrically, the removable directions are the intersection of the redundant space and the allowed orbital-variation space. A redundant vector cannot simply be truncated to the localization region: its truncated part need not remain redundant. Equation (9) instead identifies combinations whose components outside that region cancel.

Let $\mathbf{N}_i$ have orthonormal columns spanning the null space of $\mathbf{F}_i\mathbf{G}_i$. The admissible redundant directions, expressed in the allowed coefficients, are

$$\mathrm{range}(\mathbf{N}_i) = \ker(\mathbf{F}_i\mathbf{G}_i), \qquad \mathbf{K}_i = (\mathbf{G}_i\mathbf{N}_i)_{\mathcal{A}_i}, \tag{10}$$

where the subscript $\mathcal{A}_i$ means retaining the rows indexed by the allowed AOs. For an OEO with all coefficients variable, there are no excluded rows and $\mathbf{N}_i$ can be chosen as the identity. We then construct an orthonormal complement $\mathbf{Q}_i$ satisfying

$$\mathrm{range}(\mathbf{Q}_i) = \ker(\mathbf{K}_i^T), \qquad \mathbf{Q}_i^T\mathbf{Q}_i = \mathbf{I}, \qquad r_i = |\mathcal{A}_i| - \mathrm{rank}(\mathbf{K}_i), \tag{11}$$

where r_i is the number of retained coordinates for orbital i. Thus, any allowed coefficient variation admits a decomposition

$$\delta\mathbf{x}_i = \mathbf{K}_i\boldsymbol{\alpha}_i + \mathbf{Q}_i\mathbf{z}_i, \tag{12}$$

where $\boldsymbol{\alpha}_i$ gives the redundant component and $\mathbf{z}_i$ parametrizes the complement before metric conditioning. Singular value decompositions determine both null spaces. For a matrix $\mathbf{L}$, singular values exceeding $\epsilon_{\mathrm{mach}} \max(\mathrm{rows}\,\mathbf{L}, \mathrm{cols}\,\mathbf{L})\sigma_{\max}(\mathbf{L})$ are counted toward its numerical rank, where ϵ_{mach} is machine precision. The complement is Euclidean in the coefficient coordinates; it is a choice of representatives, not an orthogonality condition on the physical orbitals.

The complement is subsequently conditioned using the change of the normalized orbital. Let $\mathbf{S}_i$ be the AO overlap matrix restricted to $\mathcal{A}_i$. The normalized coefficients and their Jacobian are

$$\begin{aligned} \rho_i &= (\mathbf{x}_i^T\mathbf{S}_i\mathbf{x}_i)^{1/2}, \quad \overline{\mathbf{x}}_i = \mathbf{x}_i/\rho_i, \\ \mathbf{J}_i &= \frac{\mathbf{I} - \overline{\mathbf{x}}_i\overline{\mathbf{x}}_i^T\mathbf{S}_i}{\rho_i}, \quad \delta\overline{\mathbf{x}}_i = \mathbf{J}_i\delta\mathbf{x}_i, \end{aligned} \tag{13}$$

where ρ_i is the orbital norm and $\mathbf{J}_i$ is the derivative of normalization. In particular, $\mathbf{J}_i\mathbf{x}_i = \mathbf{0}$, which explicitly removes a radial scaling variation. On the complement, define

$$\mathbf{B}_i = \mathbf{Q}_i^T \mathbf{J}_i^T \mathbf{S}_i \mathbf{J}_i \mathbf{Q}_i, \qquad \widehat{\mathbf{U}}_i = \mathbf{Q}_i \mathbf{B}_i^{-1/2}. \tag{14}$$

Here $\mathbf{B}_i$ measures the AO-metric lengths of normalized orbital variations. Its inverse square root rescales those variations to unit metric. The inactive and active Gram matrices used above, and $\mathbf{B}_i$, must be numerically positive definite on their retained spaces; rank-deficient cases are rejected.

For HAO and selected-structure calculations, when orbital scaling is an admissible variation, the final representative is chosen tangent to the normalization sphere:

$$\mathbf{U}_i = \rho_i \mathbf{J}_i \widehat{\mathbf{U}}_i, \qquad \mathbf{x}_i^T \mathbf{S}_i \mathbf{U}_i = \mathbf{0}. \tag{15}$$

This changes $\widehat{\mathbf{U}}_i$ only by a radial, redundant component. For a complete OEO active-space expansion, the implementation instead retains $\mathbf{U}_i = \widehat{\mathbf{U}}_i$, preserving the representative of the coupled inactive and active basis redundancies. The same choice is made when fixed coefficients prevent an admissible radial adjustment. In either case,

$$(\mathbf{J}_i \mathbf{U}_i)^T \mathbf{S}_i (\mathbf{J}_i \mathbf{U}_i) = \mathbf{I}. \tag{16}$$

This is local normalization-metric conditioning, not a replacement for the full physical trust-region metric introduced below.

The formulas above assume that all coefficients in each localization region are variable. If some stored coefficients are fixed, the construction of $\mathbf{F}_i$ treats their variations as forbidden, while orbital norms and normalization derivatives include their current values. The complement is extended by zero rows on those fixed coefficients before applying $\mathbf{J}_i$; only its variable rows are retained in the final map. This preserves the same construction without allowing fixed coefficients to move.

Collecting the orbital maps gives the global nonredundant update,

$$\begin{aligned} \mathbf{U}_k &= \operatorname{diag}(\mathbf{U}_1, \mathbf{U}_2, \ldots), & \mathbf{p} &= (\mathbf{p}_1^T, \mathbf{p}_2^T, \ldots)^T, \\ \mathbf{x}(\mathbf{p}) &= \mathbf{x}_k + \mathbf{U}_k \mathbf{p}, & \boldsymbol{\theta}(\mathbf{p}) &= \mathcal{P}(\mathbf{x}_k + \mathbf{U}_k \mathbf{p}), \end{aligned} \tag{17}$$

where $\mathbf{x}_k$ is the vector of variable AO coefficients at accepted iteration k and $\mathbf{p}_i$ contains the r_i metric-conditioned coordinates of orbital i, giving a total dimension $\sum_i r_i$. The map $\mathcal{P}$ normalizes the orbitals and applies the inactive-space projection and admissible representative choices used in evaluating the wave function; $\boldsymbol{\theta}$ denotes the resulting orbital coefficients. The map $\mathbf{U}_k$ is fixed while solving the local optimization problem and rebuilt after an accepted step.

2.3 Coupled Response

The response derivation assumes that $\mathbf{M}$ is positive definite on the working structure space and that the averaged states are separated from the remaining states by an energy gap. Degeneracies within the averaged manifold are allowed.

For an orbital direction, let a dot denote the derivative at the current accepted point. Stationarity with respect to the structure coefficients gives

$$\dot{\mathcal{E}} = \frac{1}{N_{\text{state}}} \sum_{n=1}^{N_{\text{state}}} \mathbf{C}_n^T (\dot{\mathbf{H}} - E_n \dot{\mathbf{M}}) \mathbf{C}_n. \tag{18}$$

Here $\dot{\mathbf{H}}$ and $\dot{\mathbf{M}}$ are the directional derivatives of the structure Hamiltonian and overlap matrices, and $\dot{\mathcal{E}}$ is the corresponding energy derivative. Structure response first enters the second derivative.

Derivatives in the reduced coordinates follow from the orbital map in eq (17). For a scalar function F of prepared orbitals, the second derivative is

$$D^2(F \circ \boldsymbol{\theta})[\mathbf{u}, \mathbf{v}] = D^2F[D\boldsymbol{\theta}\mathbf{u}, D\boldsymbol{\theta}\mathbf{v}] + DF[D^2\boldsymbol{\theta}[\mathbf{u}, \mathbf{v}]], \tag{19}$$

where D and D^2 denote first and second differentials, $\mathbf{u}$ and $\mathbf{v}$ are reduced orbital directions, and all differentials are evaluated at $\mathbf{p} = 0$. The preparation-map contribution is included in the orbital curvature action.

Collect the selected eigenvectors in $\mathbf{C}_{\text{sel}} = (\mathbf{C}_1, \ldots, \mathbf{C}_{N_{\text{state}}})$. Differentiation of $\mathbf{C}_{\text{sel}}^T\mathbf{M}\mathbf{C}_{\text{sel}} = \mathbf{I}$ fixes the symmetric selected-space component of their response. We choose the internal gauge

$$\dot{\mathbf{C}}_{\text{sel},g} = -\tfrac{1}{2}\mathbf{C}_{\text{sel}}(\mathbf{C}_{\text{sel}}^T\dot{\mathbf{M}}\mathbf{C}_{\text{sel}}), \qquad \dot{\mathbf{C}}_n = \dot{\mathbf{C}}_{g,n} + \frac{\mathbf{Z}\mathbf{q}_n}{d_n}, \quad d_n = \sqrt{2/N_{\text{state}}}. \tag{20}$$

Here $\dot{\mathbf{C}}_{\text{sel},g}$ is the selected-space response required by normalization, $\dot{\mathbf{C}}_{g,n}$ is its column for state n, and $\mathbf{q}_n$ contains the independent response coordinates for that state. The factor d_n accounts for the equal state weight. The matrix $\mathbf{Z}$ spans the complement of the selected states and satisfies $\mathbf{C}_{\text{sel}}^T\mathbf{M}\mathbf{Z} = 0$ and $\mathbf{Z}^T\mathbf{Z} = \mathbf{I}$. Rotations within the selected equally weighted eigenspace leave the averaged energy unchanged and are excluded from the combined response vector $\mathbf{q} = (\mathbf{q}_1^T, \ldots, \mathbf{q}_{N_{\text{state}}}^T)^T$.

Writing $\dot{\mathbf{H}}[\mathbf{p}]$ and $\dot{\mathbf{M}}[\mathbf{p}]$ for orbital directional derivatives, differentiation of the generalized eigenproblem and projection with $\mathbf{Z}^T$ give

$$\begin{aligned}(\mathcal{H}^{ss}[\mathbf{q}])_n &= \mathbf{Z}^T(\mathbf{H} - E_n\mathbf{M})\mathbf{Z}\mathbf{q}_n, \\ (\mathcal{H}^{so}[\mathbf{p}])_n &= d_n\mathbf{Z}^T\left[(\dot{\mathbf{H}}[\mathbf{p}] - E_n\dot{\mathbf{M}}[\mathbf{p}])\mathbf{C}_n + (\mathbf{H} - E_n\mathbf{M})\dot{\mathbf{C}}_{g,n}[\mathbf{p}]\right],\end{aligned} \tag{21}$$

where $\mathcal{H}^{ss}$ is the structure curvature operator and $\mathcal{H}^{so}$ maps an orbital direction to a structure-response forcing. The subscript n extracts the response for state n. The first action is evaluated through Hamiltonian and overlap actions on $\mathbf{Z}\mathbf{q}_n$. The derivative of the eigenvalue drops out because $\mathbf{Z}^T\mathbf{M}\mathbf{C}_n = 0$. The factors d_n absorb the state weights and the factor of two in the real-wavefunction coefficient Hessian. The gauge term in eq (21) is essential for a moving nonorthogonal structure metric.

Let $\mathcal{H}^{oo}[\mathbf{v}]$ denote the orbital Hessian–vector product (HVP) after enforcing this selected-space normalization gauge and the orbital-map pullback, but before eliminating horizontal structure response. To define its bilinear form, introduce $\mathbf{K}_n = \mathbf{H} - E_n\mathbf{M}$, $\mathbf{K}_{n,u} = D\mathbf{H}[\mathbf{u}] - E_nD\mathbf{M}[\mathbf{u}]$, and $\mathbf{K}_{n,uv} = D^2\mathbf{H}[\mathbf{u}, \mathbf{v}] - E_nD^2\mathbf{M}[\mathbf{u}, \mathbf{v}]$, with E_n held at its accepted-point value in these definitions. The second variation of the normalization-constrained energy gives

$$\begin{aligned}\mathbf{u}^T\mathcal{H}^{oo}[\mathbf{v}] = \frac{1}{N_{\text{state}}}\sum_n \big\{&\mathbf{C}_n^T\mathbf{K}_{n,uv}\mathbf{C}_n + 2\dot{\mathbf{C}}_{g,n}[\mathbf{u}]^T\mathbf{K}_{n,v}\mathbf{C}_n \\ &+ 2\dot{\mathbf{C}}_{g,n}[\mathbf{v}]^T\mathbf{K}_{n,u}\mathbf{C}_n + 2\dot{\mathbf{C}}_{g,n}[\mathbf{u}]^T\mathbf{K}_n\dot{\mathbf{C}}_{g,n}[\mathbf{v}]\big\}.\end{aligned} \tag{22}$$

Here $\mathbf{u}^T\mathcal{H}^{oo}[\mathbf{v}]$ is the bilinear second variation in orbital directions $\mathbf{u}$ and $\mathbf{v}$. The sum runs over the N_{state} selected states. All matrix derivatives in this expression include the orbital preparation map. Second derivatives of the coefficient gauge cancel against the normalization multipliers at

a stationary structure solution; the first derivatives of that gauge remain. The constrained energy expansion is

$$Q_k(\mathbf{p},\mathbf{q}) = \mathcal{E}_k + \mathbf{g}_o^T\mathbf{p} + \tfrac{1}{2}\mathbf{p}^T\mathcal{H}^{oo}[\mathbf{p}] + \mathbf{q}^T\mathcal{H}^{so}[\mathbf{p}] + \tfrac{1}{2}\mathbf{q}^T\mathcal{H}^{ss}[\mathbf{q}], \tag{23}$$

where Q_k is the local quadratic energy model, $\mathcal{E}_k = \mathcal{E}(\mathbf{x}_k)$ is the accepted-point energy, and $\mathbf{g}_o$ is its reduced orbital gradient, obtained from eq (18) in the coordinates of eq (17). The reverse coupling action is denoted by $\mathcal{H}^{os}$, with $\mathbf{p}^T\mathcal{H}^{os}[\mathbf{q}] = \mathbf{q}^T\mathcal{H}^{so}[\mathbf{p}]$.

For a separated lowest-state manifold, $\mathcal{H}^{ss}$ is positive definite on the horizontal space. Define the response map $\mathcal{T}[\mathbf{f}]$ as the solution of $\mathcal{H}^{ss}[\mathcal{T}[\mathbf{f}]] = \mathbf{f}$. If this operator is singular, the minimum-norm solution is defined only for admissible forcing in its range. Eliminating structure response yields

$$\begin{aligned} \mathbf{q}(\mathbf{p}) &= -\mathcal{T}[\mathcal{H}^{so}[\mathbf{p}]], \\ \mathcal{H}_{\mathrm{rel}}[\mathbf{p}] &= \mathcal{H}^{oo}[\mathbf{p}] - \mathcal{H}^{os}[\mathcal{T}[\mathcal{H}^{so}[\mathbf{p}]]], \end{aligned} \tag{24}$$

where $\mathcal{H}_{\mathrm{rel}}[\mathbf{p}]$ is the relaxed orbital HVP, including structure-coefficient relaxation. The response map is evaluated by an iterative linear solve. Small structure gaps can make the relaxation contribution large even when the coupling action is small. A complete eigenvalue spectrum is not needed to evaluate eq (24); iterative actions on the admissible response space suffice.

2.4 Norm-Extended Optimization

The trust-region norm measures the physical orbital displacement. Let $\mathcal{L}_k$ map a packed coefficient tangent to AO-metric-weighted horizontal orbital features. For individually meaningful active orbitals, these features are the differentials of normalized, inactive-projected orbital rays, weighted by $\mathbf{S}^{1/2}$. For inactive orbitals, and for the active orbitals of a complete full-AO active space, horizontal subspace tangents replace individual-ray tangents; inverse Gram factors remove dependence on the chosen basis within each subspace. Variations of the inactive projector are included in the active features.

On the nonredundant coordinates, the metric and its dual norm are

$$\mathbf{G}_o = \mathbf{U}_k^T\mathcal{L}_k^T\mathcal{L}_k\mathbf{U}_k, \qquad \|\mathbf{p}\|_{\mathbf{G}_o}^2 = \mathbf{p}^T\mathbf{G}_o\mathbf{p}, \qquad \|\mathbf{r}_o\|_{\mathbf{G}_o^{-1}}^2 = \mathbf{r}_o^T\mathbf{G}_o^{-1}\mathbf{r}_o, \tag{25}$$

where $\mathbf{G}_o$ is the physical orbital metric and $\mathbf{r}_o$ is an orbital gradient or stationarity residual. We assume $\mathbf{G}_o$ is positive definite on the retained coordinates; its action is evaluated from the orbital features.

The coupled quadratic model is minimized within an orbital trust region:

$$\min_{\mathbf{p},\mathbf{q}} Q_k(\mathbf{p},\mathbf{q}) \quad \text{subject to} \quad \mathbf{p}^T\mathbf{G}_o\mathbf{p} \le \Delta_k^2, \tag{26}$$

where $\Delta_k > 0$ is the trust-region radius at accepted iteration k. The norm constraint acts on the orbital step. The stationary equations are

$$\begin{aligned} \mathcal{H}^{oo}[\mathbf{p}] + \mathcal{H}^{os}[\mathbf{q}] + \lambda\mathbf{G}_o\mathbf{p} &= -\mathbf{g}_o, \\ \mathcal{H}^{so}[\mathbf{p}] + \mathcal{H}^{ss}[\mathbf{q}] &= \mathbf{0}, \\ \lambda \ge 0, \qquad \lambda(\mathbf{p}^T\mathbf{G}_o\mathbf{p} - \Delta_k^2) &= 0, \end{aligned} \tag{27}$$

where λ is the nonnegative Lagrange multiplier for the orbital norm constraint. The last equality is the complementarity condition: λ can be positive only for a boundary step. Under the assumptions of eq (24), elimination gives $\mathcal{H}_{\mathrm{rel}}[\mathbf{p}] + \lambda \mathbf{G}_o \mathbf{p} = -\mathbf{g}_o$. The exact minimum additionally requires positive semidefiniteness of the shifted relaxed Hessian. A small stationarity residual alone does not certify that curvature condition.

Norm-extended optimization (NEO) connects a restricted Newton step to a gradient-scaled augmented eigenproblem [1]. In the eliminated orbital space, this relation reads

$$\begin{aligned} \alpha \mathbf{g}_o^T \mathbf{y} &= \mu\beta, \\ \alpha\beta \mathbf{g}_o + \mathcal{H}_{\mathrm{rel}}[\mathbf{y}] &= \mu \mathbf{G}_o \mathbf{y}, \end{aligned} \tag{28}$$

where α scales the gradient coupling, μ is an augmented generalized eigenvalue, and $(\beta, \mathbf{y}^T)^T$ is its eigenvector, with scalar homogeneous component β and orbital component $\mathbf{y}$. For $\alpha > 0$ and $\beta \neq 0$, the extracted step satisfies

$$\mathbf{p} = \frac{\mathbf{y}}{\alpha\beta}, \qquad \lambda = -\mu, \qquad \mathcal{H}_{\mathrm{rel}}[\mathbf{p}] + \lambda \mathbf{G}_o \mathbf{p} = -\mathbf{g}_o, \qquad \alpha^2 = \frac{\lambda}{-\mathbf{g}_o^T \mathbf{p}}. \tag{29}$$

Here the augmented eigenvalue determines the same orbital shift λ that appears in eq (27). The unshifted interior Newton step is recovered as a limiting case; a zero homogeneous component or a hard case requires direct treatment of the restricted quadratic problem rather than division by β. Our solver obtains the small projected restricted step spectrally and uses eqs (28) and (29) to express its norm-extended interpretation.

2.5 Iterative Solution

Orbital and structure spaces are expanded independently. Let $\mathbf{V}$ and $\mathbf{W}$ contain the retained orbital and structure basis directions, respectively. We write the trial response as $\mathbf{p} = \mathbf{V}\boldsymbol{\xi}$ and $\mathbf{q} = \mathbf{W}\boldsymbol{\zeta}$, where $\boldsymbol{\xi}$ and $\boldsymbol{\zeta}$ are the corresponding expansion coefficients. The projected matrices are

$$\begin{aligned} \mathbf{h}^{oo} &= \mathbf{V}^T \mathcal{H}^{oo}[\mathbf{V}], & \mathbf{h}^{so} &= \mathbf{W}^T \mathcal{H}^{so}[\mathbf{V}], \\ \mathbf{h}^{ss} &= \mathbf{W}^T \mathcal{H}^{ss}[\mathbf{W}], & \mathbf{G}_V &= \mathbf{V}^T \mathbf{G}_o \mathbf{V}, \\ \mathbf{h}_{\mathrm{rel}} &= \mathbf{h}^{oo} - \mathbf{h}^{os} (\mathbf{h}^{ss})^{\dagger} \mathbf{h}^{so}. \end{aligned} \tag{30}$$

where $\mathbf{h}^{oo}$, $\mathbf{h}^{so}$, $\mathbf{h}^{ss}$, and $\mathbf{G}_V$ are small subspace matrices, $\mathbf{h}^{os} = (\mathbf{h}^{so})^T$, and $\mathbf{h}_{\mathrm{rel}}$ is the projected relaxed curvature matrix. An operator acting on a basis matrix applies independently to its columns. Curvature and coupling are evaluated by matrix-free actions, with only their small subspace projections stored as matrices. The dagger on the small matrix denotes its Moore–Penrose pseudoinverse. New directions require new operator images; previously computed images are retained at the same accepted orbital point. Changing the trust radius does not invalidate these images. Full-space residuals determine whether the subspaces must be enlarged.

Certification uses residuals in the original retained coordinate spaces:

$$\begin{aligned} \mathbf{r}_o &= \mathbf{g}_o + \mathcal{H}^{oo}[\mathbf{p}] + \mathcal{H}^{os}[\mathbf{q}] + \lambda \mathbf{G}_o \mathbf{p}, \\ \mathbf{r}_s &= \mathcal{H}^{so}[\mathbf{p}] + \mathcal{H}^{ss}[\mathbf{q}], \\ \left(\|\mathbf{r}_o\|^2_{\mathbf{G}_o^{-1}} + \|\mathbf{r}_s\|_2^2 \right)^{1/2} &\leq \epsilon_{\mathrm{abs}} + \eta_k \|\mathbf{g}_o\|_{\mathbf{G}_o^{-1}}, \end{aligned} \tag{31}$$

where $\mathbf{r}_o$ and $\mathbf{r}_s$ are the orbital and structure stationarity residuals, respectively, $\epsilon_{\text{abs}} \geq 0$ is an absolute residual tolerance, and $\eta_k \geq 0$ is the iteration-dependent relative forcing tolerance. The notation $\|\cdot\|_2$ denotes the Euclidean norm; for the structure residual it refers specifically to the scaled coordinates in eq (20). Step feasibility and complementarity are checked separately. Curvature information from the projected space is not a general certificate of positive semidefiniteness outside that space.

A small structure residual can still give a significant orbital error when $\mathcal{H}^{ss}$ is ill-conditioned. Where needed, we solve for a response correction and examine its orbital effect:

$$\mathcal{H}^{ss}[\delta\mathbf{q}] = -\mathbf{r}_s, \qquad \mathbf{r}_o^{\text{corr}} = \mathbf{r}_o + \mathcal{H}^{os}[\delta\mathbf{q}]. \tag{32}$$

Here $\delta\mathbf{q}$ is a correction to the candidate structure response and $\mathbf{r}_o^{\text{corr}}$ is the orbital residual after that correction. An unresolved correction supplies an additional response direction; a sufficiently accurate correction completes the candidate response. This targets the response needed by the current orbital candidate instead of solving every orbital basis direction to final response accuracy. Appending an orbital direction requires its orbital-to-structure image.

2.6 Step Acceptance and Convergence

After applying the orbital step in eq (17), the structure coefficients are reoptimized at the trial orbitals. Acceptance compares the actual relaxed energy decrease with the coupled-model prediction,

$$\rho_k = \frac{\mathcal{E}_k - \mathcal{E}(\mathbf{x}_k + \mathbf{U}_k\mathbf{p})}{Q_k(\mathbf{0},\mathbf{0}) - Q_k(\mathbf{p},\mathbf{q})}, \tag{33}$$

where ρ_k is the ratio of actual to predicted energy decrease. The numerator uses the reoptimized structure energy at the trial orbitals, whereas the denominator uses the local quadratic model. Accepted points receive a new true gradient, chart, and local model. Rejected trials retain the accepted-point operator data while the trust radius is adjusted. A previous orbital direction may provide a warm start; the curvature model is rebuilt at each accepted point.

The full coupled model must be distinguished from the production core-only approximation. The latter retains direct inactive/core response and the nonlinear pullback with fixed upstream adjoints, but omits both local active response and horizontal structure response. The current admission decision compares response-space dimension and estimated contraction work with orbital-space dimension and core work. This is a computational work heuristic, not an error bound on omitted curvature. In both branches, trial energies and accepted gradients are evaluated for the same relaxed VBSCF objective. Full and core-only calculations must be identified separately in the numerical results.

Near a smooth isolated minimum with a nonsingular quotient Hessian, the full model approaches an ordinary Newton step once the trust region is inactive. Superlinear local convergence requires the effective Newton residual to be $o(\|\mathbf{g}_o\|)$; quadratic convergence requires an $O(\|\mathbf{g}_o\|^2)$ residual, locally Lipschitz second derivatives, and correspondingly controlled structure-solver and derivative errors. A forcing sequence $\eta_k = O(\|\mathbf{g}_o\|)$ supplies the relative part of this requirement, but a fixed nonzero ϵ_{abs} eventually limits the certified accuracy. These local conclusions do not automatically apply to the core-only model or establish rapid entry into the Newton neighborhood.

The iterative solver uses a one-electron block preconditioner. The comparison methods are block-preconditioned L-BFGS (block-LBFGS), which retains secant history with a block initial

inverse, and truncated Newton with Hessian–vector products (TNHVP), which adds conditional HVP-based corrections to the block-LBFGS step.

3 IMPLEMENTATION

3.1 Derivative Actions

Describe the accepted-point data required for gradient and HVP evaluation: normalization, inactive projection, integral transformation, structure action, and response. Identify reused quantities and directional quantities, without claiming that matrix-free means no large intermediates. Give memory costs in orbital, structure, and unique-string dimensions.

3.2 Model Updates

Give the acceptance and trust-radius update settings associated with eq (33). State which quantities are refreshed at an accepted point and which are reused during subspace expansion. Count rejected trials as work even when they do not advance the accepted-iteration counter.

Algorithm 1. Accepted-point second-order VBSCF optimization (outline)

1. Read the same orbital guess and structure list used by all comparison methods.
2. Solve the selected structure problem; evaluate energy and the physical orbital gradient.
3. Build the support-preserving local coordinates, metric, derivative actions, and preconditioner.
4. Expand the iterative second-order model; certify its residual and record response treatment.
5. Retract the orbital step, solve the trial structure problem, and evaluate model agreement.
6. Accept or reject the trial; refresh accepted-point data and test both stopping criteria.

3.3 Cost accounting

Report orbital HVPs, orbital-to-structure and transpose-coupling actions, structure Hamiltonian/overlap actions, trial evaluations, and accepted steps separately. Include setup, final observables, and total process memory in end-to-end measurements; isolate optimizer time in an additional breakdown.

4 COMPUTATIONAL DETAILS

4.1 Methods and Settings

Method	Selector	Role
XMVB-style L-BFGS	ISCF=5	Conventional quasi-Newton baseline
TNHVP	ISCF=7	Matrix-free truncated Newton comparison

Method	Selector	Role
Block-LBFGS	ISCF=8	Enhanced baseline with one-electron block initial inverse
NEO	ISCF=9	Principal norm-extended second-order method

Table 1. Comparison methods. Input selectors are implementation details, not method names.

Use identical initial orbitals, geometry, basis, active electrons/orbitals, spin, selected structure list, state weights, integral representation, and final convergence tests. Compare exact integrals and the resolution-of-the-identity (RI) approximation only in separate matched groups. Dense and Davidson structure solvers must solve the same selected generalized eigenproblem to controlled accuracy.

Record the projected-gradient convention and the physical dual-metric norm. The working energy-change and projected-gradient targets are 10^{-7} hartree and 10^{-3}, respectively; confirm the precise implemented norms and final settings before collection. Use an independently tighter calculation for endpoint comparisons.

4.2 Test-set design

Group	Examples	Purpose
Small derivative oracles	F_2; small selected-structure cases	Finite differences, dense Hessian, endpoint checks
Localized HAO	241 (benzene), 240, 7975, 7963, YAMSAI	Support constraints and convergence variation
Transition metals	MnF_2 and $FeCl_2$	Conditioning, spin, and OEO/HAO behavior
Larger response spaces	CERRAS and LOFLEA	Response cost and memory
Equal-weight state average	Small multistate cases; selected pyrrole/OEO cases	Root-space and response stability

Table 2. Proposed test groups; molecular specifications and eligibility remain to be frozen.

Do not include unresolved endpoint discrepancies as successful validation. A PySCF CASSCF comparison is a variational-space reference only when the OEO structure expansion, spin/symmetry sector, integrals, and core convention match; otherwise it is a diagnostic comparison.

4.3 Hardware and reproducibility

Collect publication timings on Hanhai25 compute nodes through Slurm. Report the central processing unit (CPU) model, allocated cores, affinity, OpenMP and Basic Linear Algebra Subprograms (BLAS) thread counts, compiler flags, dependency versions, and commit identifier. Record peak resident set size (RSS) at the job/process level. Preserve nonconverged runs and resource failures in the dataset rather than dropping them.

5 RESULTS AND DISCUSSION

5.1 Derivative and response validation

Table 3 will report gradient directional differences, HVP directional differences, bilinear symmetry defects, and coupled-versus-Schur agreement on small cases. Include nonorthogonal well-conditioned and near-singular orbital points, HAO and OEO, and at least one equal-weight state average. [Insert verified numerical errors.]

5.2 Local convergence

Figure 2 will show energy error and physical gradient norm for identical controlled perturbations around a common stationary point. Establish the local convergence regime separately from the number of iterations required to enter it. [Insert trajectories and the residual forcing sequence.]

5.3 End-to-end comparison

Figures 3 and 4 will plot accepted steps and total wall time against a common system ordering, with failures visibly marked. Table 4 will provide final energies, energy changes, physical gradients, action counts, wall time, and RSS. Discuss cases where fewer steps do not yield lower wall time. [Insert matched results; do not mix revisions.]

5.4 Cost Breakdown

Figure 5 will break down setup, core HVP, coupling/structure response, trial evaluation, and final output. Analyze larger active spaces using measured work and memory rather than accepted-step count alone. Compare full and restricted response with the model treatment stated for every run. [Insert stage timings and response statistics.]

5.5 Ablations and limitations

Separate model changes from preconditioner changes. Prioritize matched full/restricted response and one-electron block/Hessian-diagonal comparisons; identity preconditioning need not be a production benchmark. Discuss small structure gaps, incomplete selected-VB spaces, root changes, and redundant-coordinate conditioning. [Insert only ablations that answer a specific algorithmic question.]

6 CONCLUSIONS

Summarize the validated second-order formulation, the measured range of efficiency, and the cost of structure relaxation. State which method is recommended for which workload only if the data support that distinction. [Write the quantitative conclusion after Sections 5.1–5.5 are complete.]

ASSOCIATED CONTENT

Supporting Information

Planned contents of the Supporting Information (SI): coordinate and response derivations; update-map derivatives; finite-difference protocols; complete molecular specifications and initial orbitals; raw convergence trajectories; nonconvergence records; timing scripts and environment details. Submit the SI separately from the main manuscript.

AUTHOR INFORMATION

Corresponding Author: Chen Zhou. Email: czhou1989@xmu.edu.cn. Author Contributions: [To be agreed by all authors]. Notes: [Competing-interest statement to be confirmed].

ACKNOWLEDGMENTS

[Funding, computing resources, and individual acknowledgments to be confirmed. Add any required AI-assistance disclosure after review of the applicable ACS policy; do not list an AI tool as an author.]

DATA AVAILABILITY STATEMENT

[Supply a persistent public repository link or DOI for the input files, initial orbitals, raw outputs, benchmark metadata, and analysis scripts. Identify the code release and commit used for every figure. Explain any justified restrictions.]

REFERENCES — STARTER LIST

(1) Jensen, H.-J. Aa.; Jørgensen, P. A Direct Approach to Second-Order MCSCF Calculations Using a Norm Extended Optimization Scheme. *J. Chem. Phys.* **1984**, *80* (3), 1204–1214.

(2) Werner, H.-J.; Knowles, P. J. A Second Order Multiconfiguration SCF Procedure with Optimum Convergence. *J. Chem. Phys.* **1985**, *82* (11), 5053–5063.

(3) Sun, Q.; Yang, J.; Chan, G. K.-L. A General Second Order Complete Active Space Self-Consistent-Field Solver for Large-Scale Systems. *Chem. Phys. Lett.* **2017**, *683*, 291–299.

[Complete the VBSCF/XMVB, inexact-Newton, trust-region, L-BFGS, and density-fitting literature before submission. This starter list is not an exhaustive priority review.]